Crime Rate Pattern Analysis

COURSE : DATA ANALYTICS

Submitted by : Jayashree K Sunagar

ABSTRACT

Crime analysis plays a vital role in understanding public safety trends and assisting authorities in effective decision-making. This project focuses on analyzing crime data across various states and union territories in India to identify crime patterns, regional variations, and future trends using data analytics and machine learning techniques.

The dataset includes crime statistics from the years 2020, 2021, and 2022 along with population data, crime rate, and chargesheeting rate. Data preprocessing and feature engineering techniques were applied to generate useful attributes such as average crime, crime growth, and crime per population. Exploratory Data Analysis (EDA) was conducted to identify crime trends, compare regional crime levels, and study relationships between crime and population.

Machine learning techniques were implemented using the Linear Regression model to predict future crime trends and estimate crime counts for the year 2023. The model performance was evaluated using metrics such as Mean Squared Error (MSE) and R-Squared Score (R^2).

The results demonstrate that data-driven approaches can effectively analyze crime patterns, support law enforcement planning, and assist in crime prevention strategies through predictive insights and interactive dashboards.

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1. INTRODUCTION

Crime is one of the major concerns affecting societal development and public safety. Monitoring and analyzing crime statistics helps governments and law enforcement agencies understand crime patterns, identify high-risk regions, and implement preventive measures. With the increasing availability of public crime data, data analytics provides an effective approach for understanding crime behaviour and making informed decisions.

This project focuses on analyzing crime data from different states and union territories across India using statistical analysis, visualization, SQL querying, and predictive modelling techniques. The objective is to identify trends in crime rates over multiple years, compare regional crime statistics, and forecast future crime trends.

Exploratory Data Analysis is used to understand crime distributions, state-wise comparisons, and chargesheeting efficiency. Linear Regression is applied to predict future crime counts based on historical data, providing insights into possible future crime trends. Overall, this project aims to provide meaningful crime insights through data-driven techniques and interactive dashboard visualization.

❖ PROJECT BACKGROUND

With growing urbanization and population, crime incidents have become a major challenge in maintaining public safety. Government organizations release annual crime statistics to track law and order conditions across states. However, manually analyzing large volumes of crime data is difficult and time-consuming.

Using data analytics and predictive modelling helps in efficiently analyzing historical crime records, identifying trends, and forecasting future crime possibilities. This project uses publicly available crime data to perform detailed analysis and prediction.

❖ PROBLEM STATEMENT

Analyzing crime trends across multiple states and years is challenging due to the large volume of statistical data and varying crime rates across regions.

This project aims to analyze historical crime data, identify crime patterns, and predict future crime trends using machine learning and data visualization techniques.

❖ **OBJECTIVES**

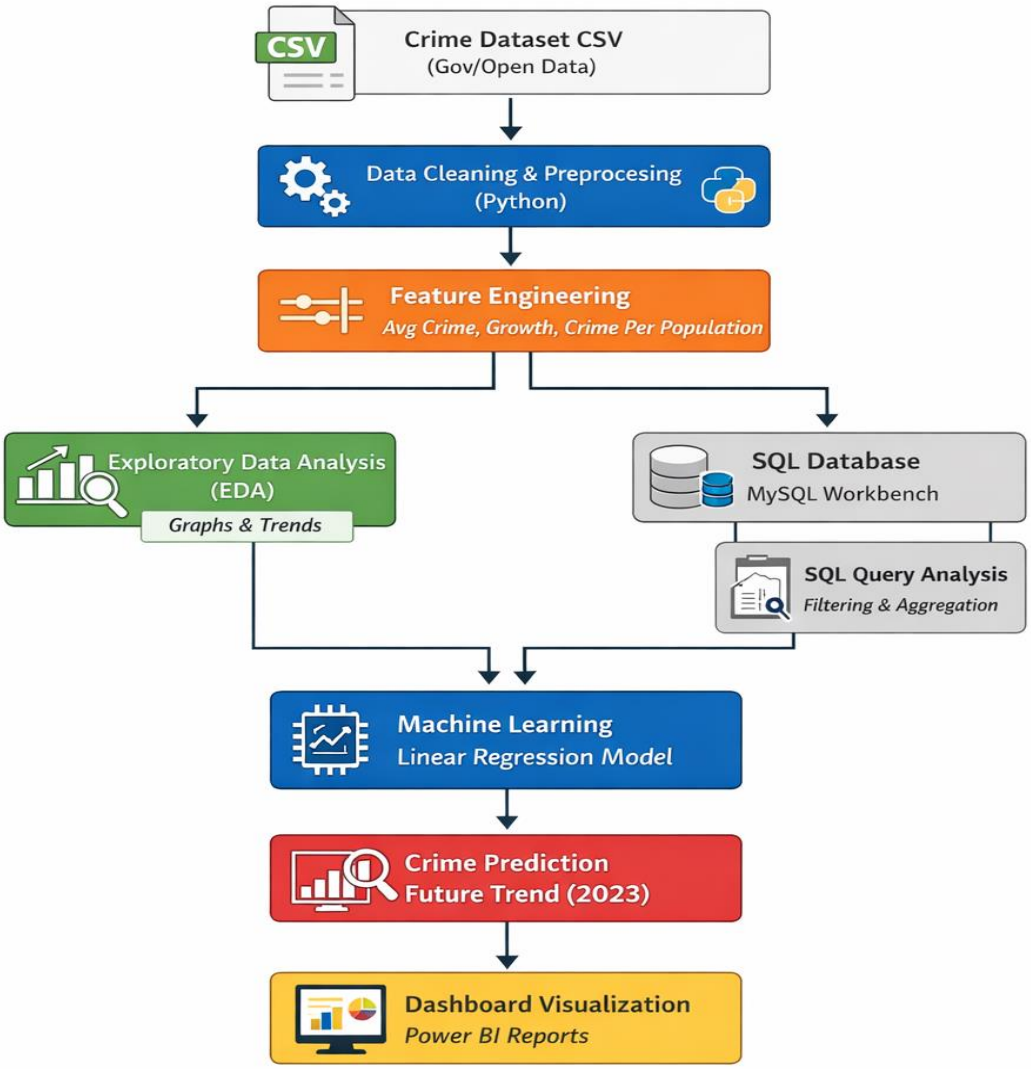
- To analyze crime statistics across multiple states and years
- To perform data preprocessing and feature engineering for improved analysis
- To identify high-crime and low-crime regions through exploratory analysis
- To predict future crime trends using Linear Regression
- To evaluate model performance using regression metrics
- To visualize crime insights using an interactive Power BI dashboard

2. TOOLS AND TECHNOLOGIES

This project uses various tools and technologies for analysis, modelling, and visualization:

- **Python** – Used for data preprocessing, analysis, and machine learning
- **Pandas, NumPy** – Used for data cleaning and numerical computations
- **Scikit-learn** – Used to implement Linear Regression model
- **Matplotlib & Seaborn** – Used for data visualization and EDA
- **MySQL** – Used for SQL-based data analysis
- **Power BI** – Used for dashboard creation and visualization

3. SYSTEM ARCHITETURE



4. DATA DESCRIPTION

Dataset Overview

The dataset used in this project contains crime statistics of Indian states and union territories from 2020 to 2022.

Dataset Features

- State/UT – Name of the state or union territory
- Crime_2020 – Total crime count in 2020
- Crime_2021 – Total crime count in 2021
- Crime_2022 – Total crime count in 2022
- Population_2022 – Mid-year projected population
- Crime_Rate_2022 – Rate of cognizable crimes in 2022
- Chargesheet_Rate_2022 – Chargesheeting rate in 2022
- Avg_Crime – Average crime across 2020–2022
- Crime_Growth – Difference between crime in 2022 and 2020
- Crime_Per_Population – Crime relative to population size

5. DATA PREPROCESSING AND FEATURE ENGINEERING

Data Preprocessing

- Removed unnecessary summary rows such as Total State and Total India
- Checked for missing/null values
- Converted numeric columns into appropriate data types
- Renamed columns for better readability

Feature Engineering

1. **Average Crime Calculation**
   Average crime is calculated using 2020, 2021, and 2022 crime values
2. **Crime Growth Calculation**
   Difference between crime in 2022 and 2020
3. **Crime Per Population**
   Ratio of crime count to total population

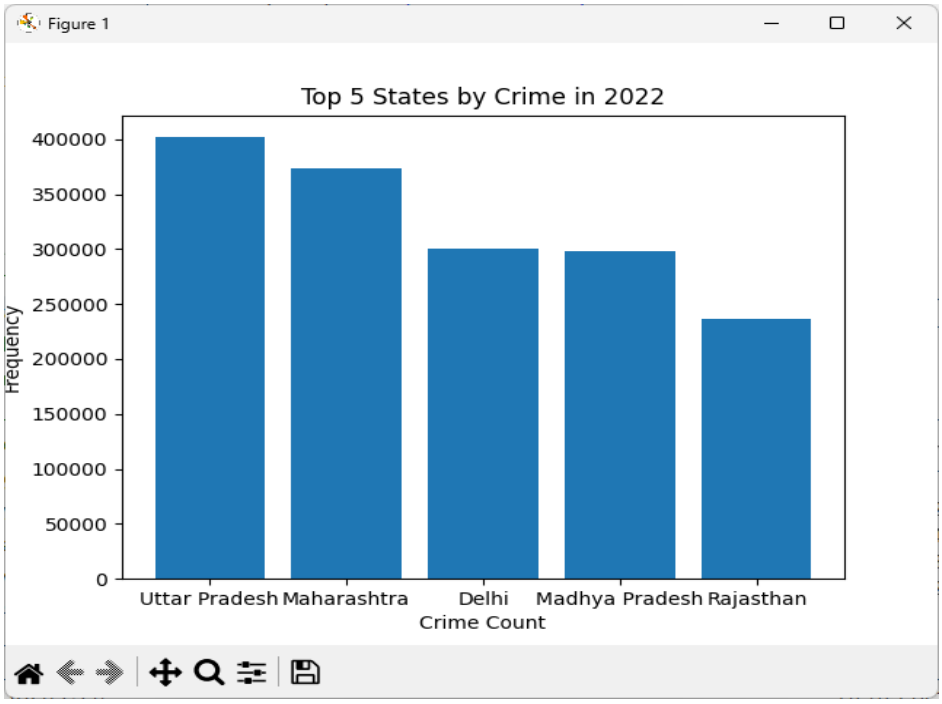
6. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was conducted to understand crime distribution, identify regional crime patterns, and analyze relationships between different crime-related factors. Various visualizations were created using Python libraries such as Matplotlib and Seaborn to derive meaningful insights from the dataset.

Top 5 States by Crime in 2022

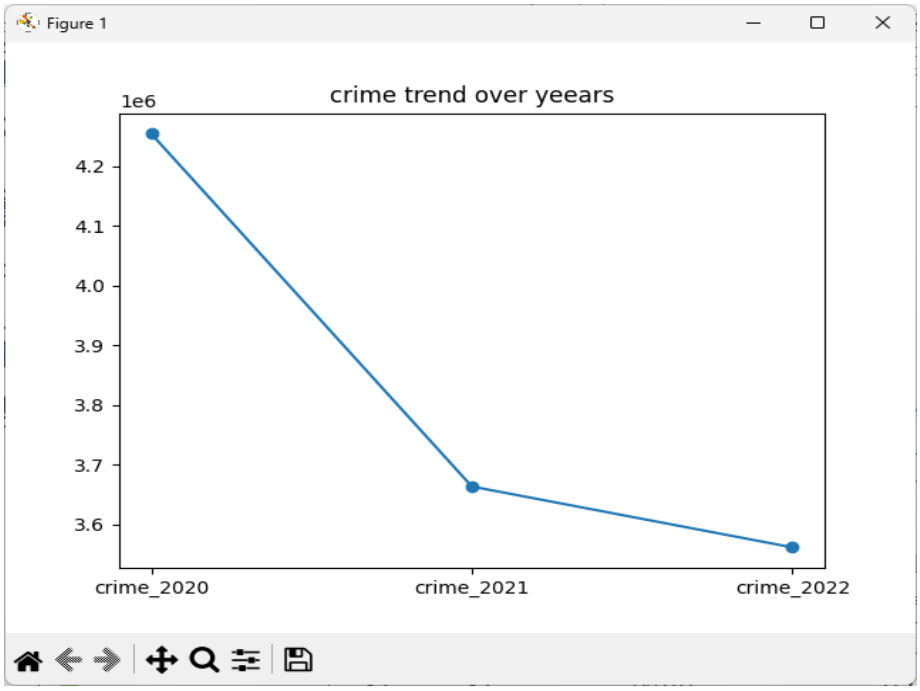
The dataset was sorted in descending order based on crime count for the year 2022 to identify the top five states/UTs with the highest reported crimes.

A bar chart was used to visualize the top 5 states with the highest crime count in 2022.



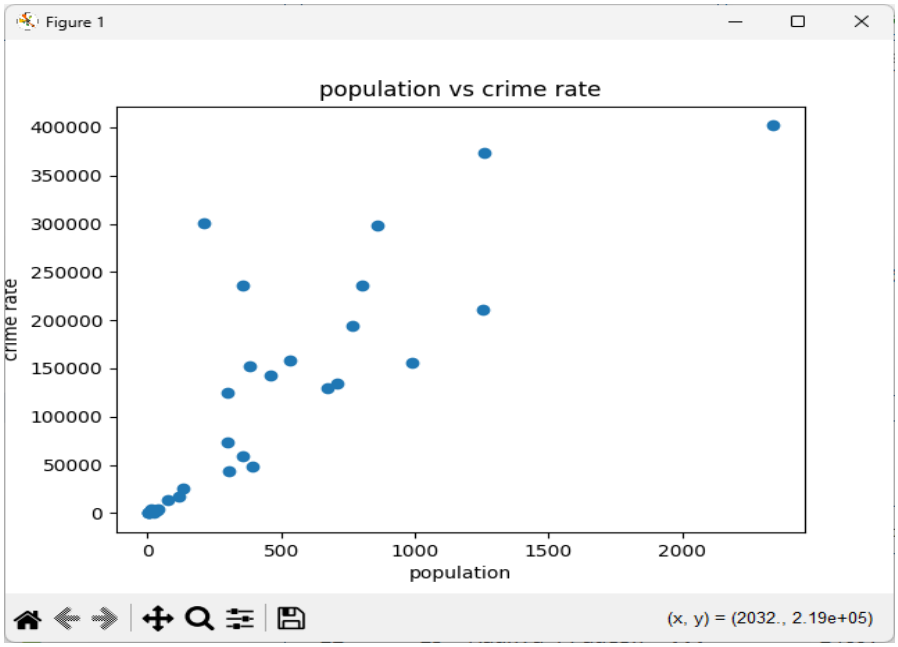
Crime Trend Over Years

The total crime counts for the years 2020, 2021, and 2022 were aggregated and plotted using a line graph.
This analysis helps in understanding the yearly crime trend and identifying whether crime has increased or decreased over time.
The line chart provides a clear overview of crime fluctuations across multiple years and supports trend-based analysis.



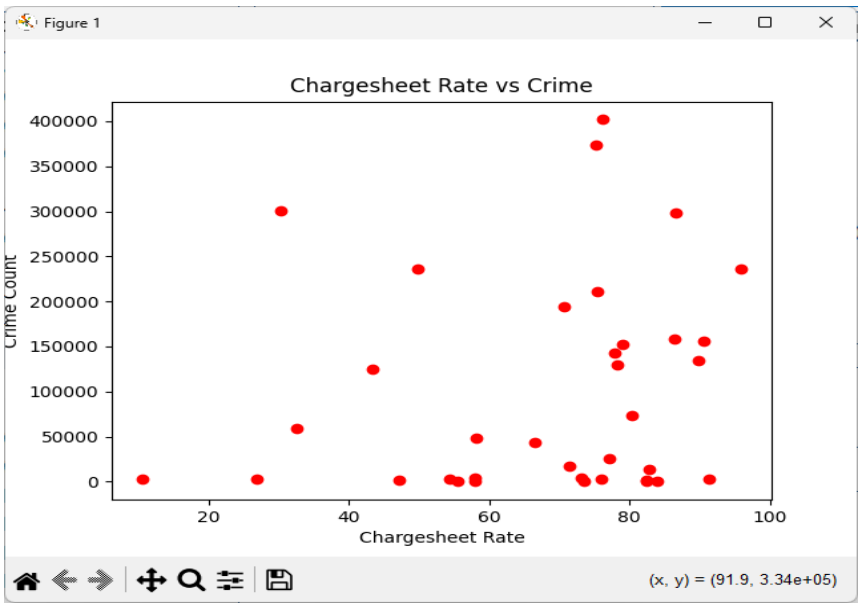
Population vs Crime Rate Analysis

A scatter plot was created to analyze the relationship between projected population and crime count in 2022.
This helps determine whether regions with larger populations tend to experience higher crime levels.The scatter plot helps identify correlation patterns between population size and crime occurrence.



Chargesheet Rate Analysis

A scatter plot was used to compare the chargesheeting rate against crime count.
This analysis evaluates the relationship between crime occurrence and law enforcement effectiveness in filing chargesheets.
It helps assess whether regions with higher crime counts also maintain proportional chargesheeting efficiency.



7. PREDICTIVE MODELLING

Predictive modelling is performed in this project to estimate future crime trends and analyze crime patterns using machine learning techniques. The **Linear Regression algorithm** is implemented to identify relationships between historical crime records and associated influencing factors, enabling prediction of future crime counts.

Linear Regression is a supervised machine learning algorithm commonly used for predicting continuous numerical values. In this project, it helps forecast crime trends based on previous crime statistics and supporting attributes.

The model is used to understand how historical crime counts, population, and chargesheeting rates influence future crime occurrences.

Feature Selection

The following input variables were selected as features for training the predictive model:

- **Crime_2020** – Represents the total crime count recorded in the year 2020
- **Crime_2021** – Represents the total crime count recorded in the year 2021
- **Population_2022** – Indicates the projected mid-year population for 2022
- **Chargesheet_Rate_2022** – Reflects the chargesheeting efficiency percentage for 2022

These features are chosen because they directly influence and contribute to predicting future crime trends.

Target Variable

The output variable used for prediction is:

- **Crime_2022** – Total crime count recorded in the year 2022

This target variable is used to train the model and evaluate prediction performance.

8. MODEL IMPLEMENTATION

Data Splitting

To build and evaluate the predictive model effectively, the dataset is divided into training and testing sets:

- **70% of the data** is used for training the model
- **30% of the data** is used for testing the model

This split ensures that the model is evaluated on unseen data to validate prediction performance.

- **Model Training**

The Linear Regression model is trained using the selected input variables and target variable.

During the training process:

- The model learns patterns and relationships between the selected features and crime counts
- Mathematical coefficients are calculated to determine the influence of each input variable on the prediction output

This enables the model to establish a regression equation for future predictions.

- **Prediction**

After training, the model is used for making predictions on testing data.

The trained model predicts:

- **Crime count for 2022** to evaluate prediction performance against actual values
- **Estimated crime count for 2023** to forecast future crime trends based on historical data

The predicted value for 2023 helps provide insights into potential future crime scenarios.

9. MODEL EVALUATION

The performance of the Linear Regression model is evaluated using regression metrics to determine prediction accuracy and reliability.

✓ **Mean Squared Error (MSE)**

Mean Squared Error measures the average squared difference between actual and predicted crime values.

It helps determine how close the predicted values are to the actual observed values.

Obtained Result:

- **MSE = 702544600.87**

A lower MSE value indicates better prediction accuracy and fewer errors in prediction.

✓ **R-Squared Score (R^2)**

R-Squared Score measures how well the independent variables explain the variation in the dependent variable.

It indicates the proportion of variance in crime count that is predictable from the selected input variables.

Obtained Result:

- **R^2 Score = 0.9693**

This means the model explains approximately **96.93% of the variance** in crime data, indicating very strong predictive capability.

✓ **Performance Interpretation**

The obtained evaluation metrics indicate that the Linear Regression model performs effectively in predicting crime trends.

- The **high R^2 score** demonstrates that the model fits the data well and captures most of the relationships between features and target variable.
- The **acceptable MSE value** indicates that prediction errors are minimal relative to the scale of the dataset.

Overall, the model provides reliable and accurate predictions for analyzing crime trends and forecasting future crime counts.

Prediction Outcome

Using the trained Linear Regression model, the predicted estimated crime count for the year **2023** is:

- **Predicted Crime Count (2023): 97,033**

This prediction helps estimate future crime levels and supports proactive planning for law enforcement and public safety management.

10. SQL-BASED ANALYSIS

Structured Query Language (SQL) is utilized in this project to manage, store, and analyze crime data efficiently using **MySQL Workbench**. SQL enables structured storage of the cleaned crime dataset in a relational database environment, allowing efficient execution of queries for analytical insights and data exploration.

The processed CSV dataset was imported into MySQL database tables through MySQL Workbench after performing data cleaning and preprocessing. Database connectivity was established successfully, and the dataset was stored in tabular format for further structured querying and analysis.

SQL-based analysis improves data handling efficiency, supports rapid querying of large datasets, and enables extraction of meaningful insights through relational database operations.

Database Connection and Data Import

The following steps were performed for SQL database setup:

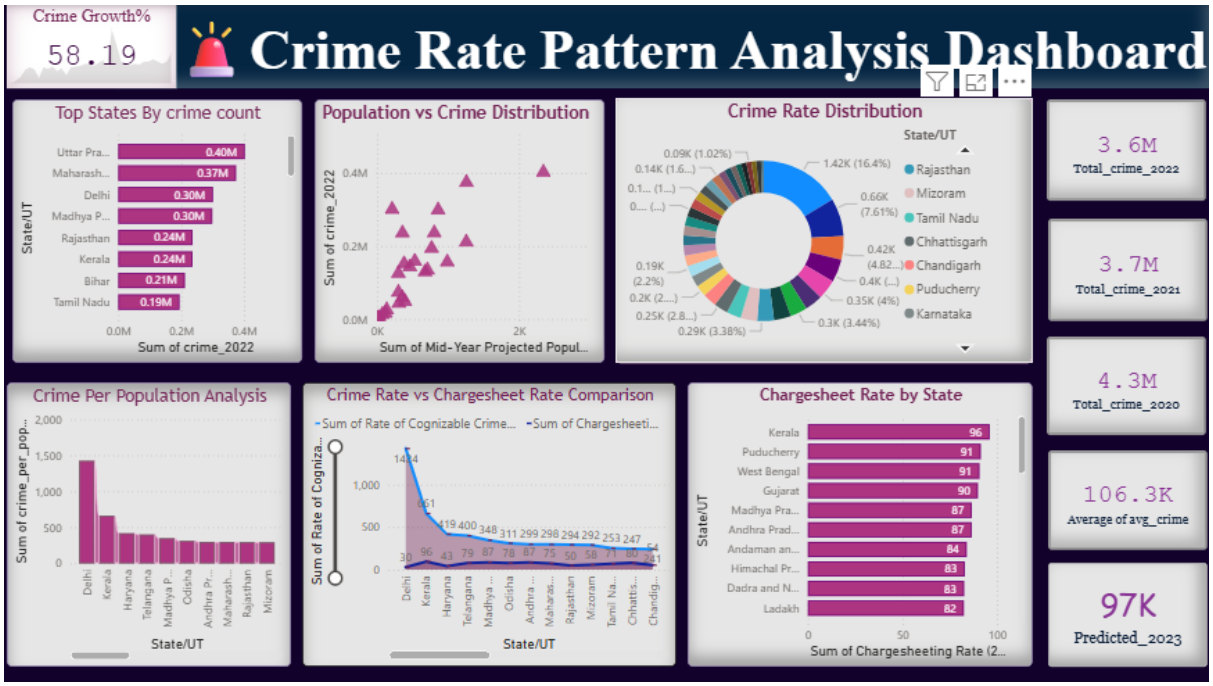
- MySQL server connection was established using **MySQL Workbench**
- A dedicated database was created for the project
- The cleaned and updated CSV crime dataset was imported into the database table
- Appropriate data types were assigned to each column during import
- SQL queries were executed to validate and analyze stored data

This setup ensures organized storage and retrieval of crime-related records for analysis.

11.DASHBOARD AND VISUALIZATION

POWER BI

The dashboard is developed using Power BI Desktop to provide interactive visualization of crime data.



It includes:

- Total Crime Statistics
- Average Crime Rate
- Chargesheet Rate Analysis
- Predicted Crime Trends
- State-wise Crime Comparison
- Crime Trend Over Years
- Population vs Crime Distribution

The dashboard helps users understand crime insights and regional patterns efficiently.

BUSINESS INSIGHTS

- Certain states consistently report higher crime rates than others
- Population growth influences crime volume in many regions
- Chargesheet rates vary significantly across states
- Crime has shown varying trends between 2020 and 2022
- Predicted crime trends help identify possible future crime patterns

CONCLUSION

In this project, crime data across Indian states and union territories was successfully analyzed using data analytics, SQL, and machine learning techniques. Feature engineering and exploratory data analysis helped in identifying important patterns such as crime growth, high-crime regions, and population-crime relationships.

A Linear Regression model was implemented to predict future crime trends, achieving strong performance with a high R^2 score. The developed dashboard provides effective visualization of crime insights, helping in better understanding and informed decision-making.

Overall, this project demonstrates how data-driven approaches can effectively analyze crime trends and support crime monitoring and prevention efforts.

FUTURE SCOPE

- Implement advanced predictive models like Random Forest and XGBoost
- Use real-time crime data for live monitoring
- Integrate geographic heatmaps for location-based crime analysis
- Build crime alert systems for high-risk regions
- Expand analysis using additional demographic and socio-economic factors

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